

Product Line Profitability Analysis for Nassau Candy Distributor

Abstract

This project analyzes the profitability of products and divisions within Nassau Candy Distributor using Python. The analysis focuses on gross profit, gross margin, revenue contribution, profit contribution, pricing inefficiencies, cost-heavy products, and margin risks. Various visualizations and dashboards were created to support business decision-making.

Objective

Identify the most profitable products.

Analyze division-wise performance.

Detect cost-heavy and low-margin products.

Measure margin volatility.

Perform Pareto (80/20) analysis.

Develop an interactive Streamlit dashboard.

Dataset

The dataset contains:

Product Name

Division

Sales

Gross Profit

Cost

Units

Order Date

Ship Date

Methodology

Data Cleaning

Removed missing values

Removed duplicate records

Removed zero-sales records

Removed negative-profit records

Converted date columns

KPI Calculations

Gross Margin %

Profit Per Unit

Revenue Contribution %

Profit Contribution %

Margin Volatility

Analysis Performed

Product Profitability Analysis

Division Performance Analysis

Cost vs Sales Analysis

Pricing Inefficiency Detection

Margin Risk Analysis

Pareto Analysis (80/20 Rule)

Key Findings

A small number of products contribute most of the profit.

Some products have high sales but low margins.

Cost-heavy products reduce profitability.

Division profitability varies significantly.

Margin-risk products require monitoring.

Dashboard Features

KPI Cards

Product Profitability Dashboard

Division Dashboard

Cost vs Sales Dashboard

Pareto Dashboard

Product Search Filter

Division Filter

Margin Filter

Conclusion

The project successfully identified profitable products, division performance, pricing inefficiencies, and margin-risk products. The Streamlit dashboard provides an interactive solution for monitoring profitability and supporting business decisions.

Tools Used

Python

Pandas

NumPy

Matplotlib

Seaborn

Streamlit